

DATA SHEET

PW15PH... Single point load cell

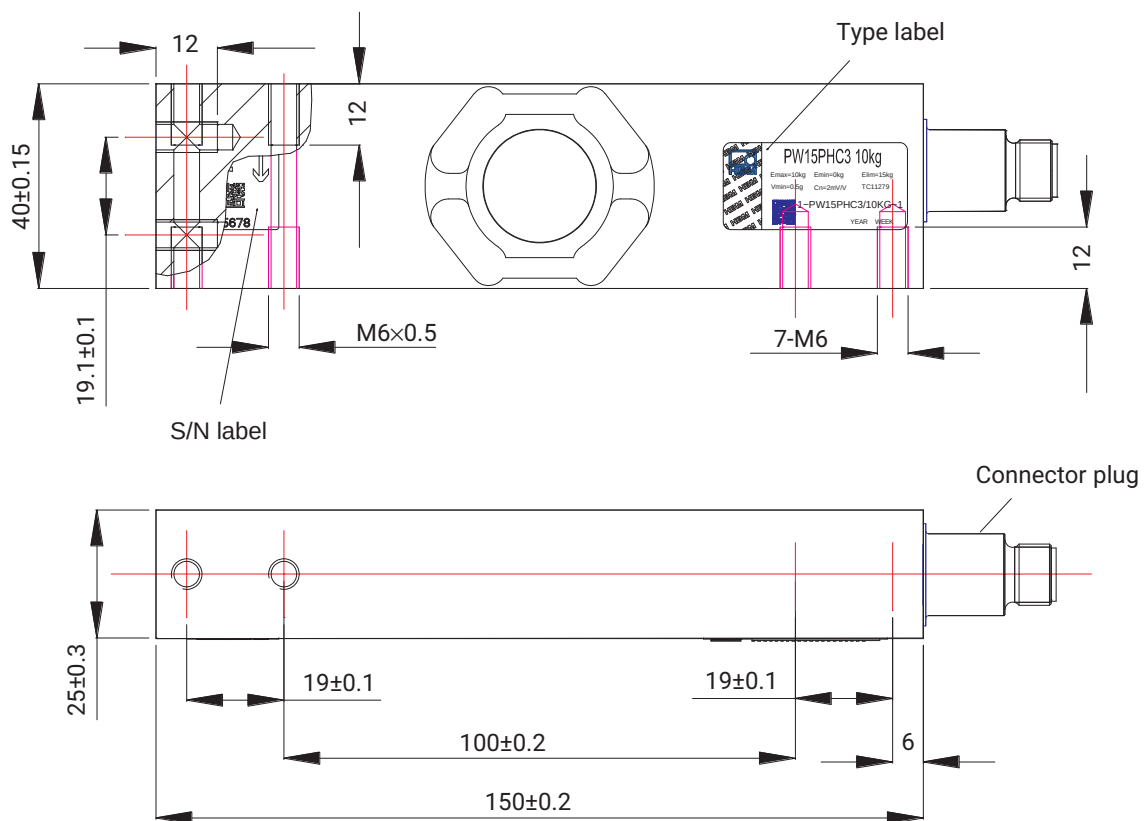
SPECIAL FEATURES

- Maximum capacities 10 ... 100 kg
- Stainless steel
- High ratio of minimum verification interval Y
- Meets EMC guidelines
- M12 connection cable (aseptic) and other options available



DIMENSIONS

Dimensions in mm; (1 mm = 0.03937 inches)



SPECIFICATIONS

Type			PW15PH (C3 MR)			
Accuracy class ¹⁾			C3 Multi Range (MR)			
Number of load cell verification intervals	n_{LC}		3,000			
Maximum capacity	E_{max}	kg	10	20	50	100
Minimum load cell verification interval	v_{min}	g	1	2	5	10
Ratio of minimum verification interval	Y		10,000			
Temperature coefficient of the zero signal per 10 K	TK_0	% of C_n / 10 K	± 0.0140			
Maximum platform size		mm	500 x 400			
Nominal (rated) sensitivity	C_n	mV/V	2.0 $\pm$ 0.2			
Zero signal error			0 $\pm$ 0.1			
Temperature coefficient of the sensitivity per 10 K ²⁾ in the temperature range +20 ... +40 °C -10 ... +20 °C	TK_C	% of C_n	± 0.0175			
Non-linearity ²⁾	d_{lin}		± 0.0117			
Relative reversibility error ²⁾	d_{hy}		± 0.0166			
Minimum dead load output return	MDLOR		± 0.0166			
Off-center load error ³⁾			± 0.0166			
Input resistance	R_{LC}	Ω	300 .. 500			
Output resistance	R_0		300 .. 500			
Reference excitation voltage	U_{ref}	V	5			
Nominal (rated) range of the excitation voltage	B_U		1 ... 12			
Maximum excitation voltage			15			
Insulation resistance at 100 V _{DC}	R_{is}	G Ω	>1			
Nominal (rated) ambient temperature range	B_T	°C	-10 ... +40			
Operating temperature range	B_{tu}		-10 ... +50			
Storage temperature range	B_{tl}		-25 ... +70			
Cleaning temperature			Max. 120 °C for max. 10 minutes			
Service load at max. 100 mm eccentricity	E_U	% of E_{max}	150			
Limit load at max. eccentricity of 160 mm	E_L		150			
Limit lateral loading, static	E_{lq}		300			
Breaking load at max. 20 mm eccentricity	E_d		300			
Relative permissible oscillation stress at max. 20 mm eccentricity	F_{srel}		70			
Nominal (rated) displacement ⁴⁾	s_{nom}	mm	<0.5			
Weight, approx.	m	kg	0.9			
Degree of protection ⁶⁾			IP68 (test conditions 1 m water column / 100 h); IP69K (water at high pressure, steam cleaner) ⁵⁾			
Measuring body material			Stainless steel 1.4545 ⁷⁾			

¹⁾ As per OIML R60, with $P_{LC} = 0.7$.

²⁾ The values for non-linearity (d_{lin}), relative reversibility error (d_{hy}) and temperature coefficient of sensitivity (TK_C) are recommended values. The sum of these values is within the cumulated error limit laid down by OIML R60.

³⁾ As per OIML R76.

⁴⁾ Loading with E_{max} and center of gravity in center of load cell.

⁵⁾ Based on DIN 40050, Part 9 specifications, for road vehicles.

⁶⁾ As per EN 60529 (IEC 529)

⁷⁾ As per EN 10088-1.

Type			PW15PH (C6 MR)			
Accuracy class ⁸⁾			C6 MR (Multi Range)			
Max. number of load cell interval	n_{LC}		6,000			
Maximum capacity ⁹⁾	E_{max}	kg	10	20	50	100
Minimum load cell verification interval	v_{min}	g	0.5	1	2	5
Ration of minimum verification interval	Y		20,000		25,000	20,000
Temperature coefficient of the zero signal	TC_0		± 0.0070		± 0.0056	± 0.0070
Temperature coefficient of the sensitivity ¹⁰⁾	TC_s	% of C_n / 10 K	± 0.0087 ± 0.0058			
Temperature range +20 ... +40 °C						
-10 ... +20 °C						
Hysteresis error ¹⁰⁾	d_{hy}	% of C_n	± 0.0083			
Non-linearity ¹⁰⁾	d_{lin}		± 0.0083			
Minimum dead load output return	MDLOR		± 0.0083			
Off-center load error ¹¹⁾			± 0.0116			

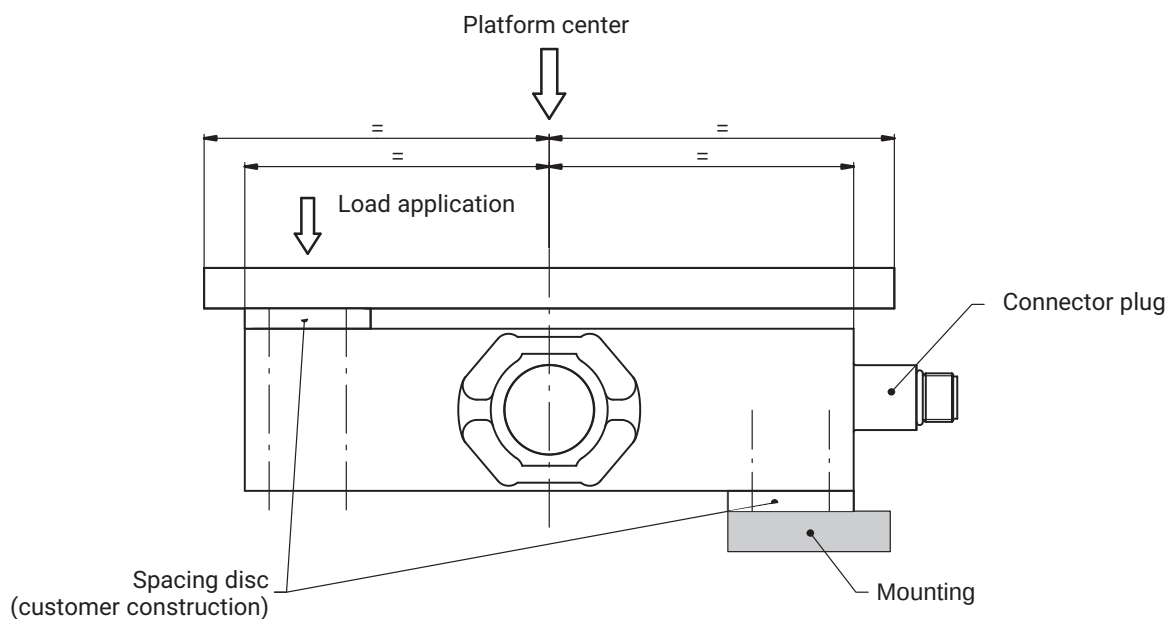
⁸⁾ As per OIML R60, with $P_{LC} = 0.7$

⁹⁾ Max. eccentric loading as per OIML R76

¹⁰⁾ The sum of data for Non-linearity, Hysteresis and TC Span meets the requirements of OIML R60

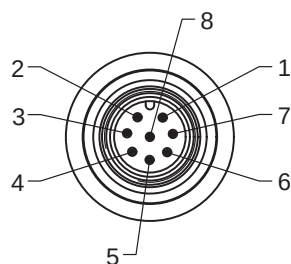
¹¹⁾ As per OIML R76

MOUNTING INSTRUCTIONS



Maximum capacities	Thread	Min. property class	Tightening torque
10 ... 100 kg	M6	10.9	14 N·m

CONNECTOR PIN ASSIGNMENT



- Plug-in contact 1 = measurement signal (+)
- Plug-in contact 2 = vacant
- Plug-in contact 3 = sense lead (+)
- Plug-in contact 4 = vacant
- Plug-in contact 5 = sense lead (-)
- Plug-in contact 6 = excitation voltage (-)
- Plug-in contact 7 = excitation voltage (+)
- Plug-in contact 8 = measurement signal (-)

PRODUCT NUMBERS (OVERVIEW)

PW15PH... (stainless steel, hermetically sealed)

Type	PW15PH	PW15PH C6-MR
Accuracy class	C3-MR (OIML) (Multi Range)	C6-MR (OIML) (Multi Range)
Comments	Plug connection	
Capacity	Order number	
10 kg	1-PW15PHC3/10KG-1	1-PW15PHC6/10KG-1
20 kg	1-PW15PHC3/20KG-1	1-PW15PHC6/20KG-1
50 kg	1-PW15PHC3/50KG-1	1-PW15PHC6/50KG-1
100 kg	1-PW15PHC3/100KG-1	1-PW15PHC6/100KG-1

ACCESSORIES

Connection cable	
Connection cable with M12 F connector, 8-pin, TPU IP67, PUR cable sheath, 5 m long	1-KAB168-5
Connection cable with M12 F connector, 8-pin, TPU IP67, PUR cable sheath, 20 m long	1-KAB168-20
Connection cable with M12 F connector, 8-pin, stainless steel IP68/IP69K, hygiene design, 3 m long	1-KAB175-3-1
Connection cable with M12 F connector, 8-pin, stainless steel IP68/IP69K, hygiene design, 6 m long	1-KAB175-6-1
Connection cable with M12 F connector, 8-pin, stainless steel IP68/IP69K, hygiene design, 12 m long	1-KAB175-12-1

For connection cable specifications, see separate data sheet B3643.

Pin assignment 1-KAB168

Color code	Connection
White	Measurement signal (+)
Red	Measurement signal (-)
Blue	Excitation voltage (+)
Pink	Excitation voltage (-)
Green	Sense lead (+)
Gray	Sense lead (-)
Yellow	Not in use
Brown	Not in use

Pin assignment 1-KAB175

Color code	Connection
White	Measurement signal (+)
Red	Measurement signal (-)
Blue	Excitation voltage (+)
Black	Excitation voltage (-)
Green	Sense lead (+)
Gray	Sense lead (-)

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